Fiscal Year:	FY 2011	**Task Last Updated:**	FY 06/15/2011
PI Name:	Tenforde, Thomas Ph.D.		
Project Title:	Impact of Individual Susceptibility and Previous Radiation Risk for Astronauts		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NON-PROFIT	**Phone:**	301-657-2652 ext. 19
Organization Name:	National Council on Radiation Protection and Measurements		
PI Address 1:	7910 Woodmont Ave		
PI Address 2:	Suite 400		
PI Web Page:	http://NCRPonline.org		
City:	Bethesda	**State:**	MD
Zip Code:	20814	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	05/01/2008	**End Date:**	03/31/2011
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Received no-cost extension to 03/31/2011 in July 2010, per NSSC information (Ed., 6/15/2011) NOTE: End date changed to 6/20/2010 in January 2010 per NSSC information; original end date 2/13/2009 (Ed., 6/13/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX08AP37G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The National Council on Radiation Protection and Measurements (NCRP) has provided guidance to the National Aeronautics and Space Administration (NASA) in several reports related to astronaut radiation protection and operational radiation safety during missions in low-Earth orbit (LEO). These reports include NCRP Report No. 98, Guidance on Radiation Received in Space Activities (1989); NCRP Report No. 132, Radiation Protection Guidance for Activities in Low-Earth Orbit (2000); and NCRP Report No. 142, Operational Radiation Safety Program for Astronauts in Low-Earth Orbit: A Basic Framework (2002). Further, NCRP has held conferences and prepared several publications on specialized topics related to NASA space activities. These include Symposium Proceedings No. 3, Acceptability of Risk from Radiation – Application to Human Space Flight (1997) and Annual Meeting Proceedings No. 20, Cosmic Radiation Exposure of Airline Crews, Passengers and Astronauts (2000). In addition, other publications of relevance to the NASA space program are NCRP Commentary No. 12, Radiation Exposure and High-Altitude Flight (1995); NCRP Report No. 131, Scientific Basis for Evaluating the Risks to Populations from Space Applications of Plutonium (2001); NCRP Report No. 137, Fluence-Based and Microdosimetric Event-Based Methods for Radiation Protection in Space (2001); and NCRP Report No. 153, Information Needed to Make Radiation Protection Recommendations for Space Missions Beyond Low-Earth Orbit (2006). NASA also contributed to NCRP Report No. 151 on Structural Shielding Design and Evaluation for Megavoltage X- and Gamma-Ray Radiotherapy Facilities (2005). The NCRP will be providing a Report to address additional questions related to the radiation protection of NASA astronauts in LEO, lunar and interplanetary missions.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	This Report addresses additional questions related to the radiation protection of NASA astronauts in LEO, lunar and interplanetary missions. The Report discusses whether genetic susceptibility factors or prior radiation exposures can influence the radiation health risks for astronauts resulting from radiation exposures that occur during space missions over the course of their lifetimes. With current technology, it is generally not possible to predict an individual's inherent genetic susceptibility to the long-term risk of cancer or other diseases associated with radiation exposure. However, it is anticipated that the results of ongoing and future medical research will lead to an ability to predict the effects of genetic susceptibility on lifetime risk of stochastic diseases. Similarly, it is expected that the influence of prior radiation exposures (e.g., from medical procedures) may be better understood as a result of current and future research on exposed populations. This Report addresses the scientific basis for projecting the impact of these important factors in establishing radiation health protection guidance to reduce the risks associated with future space missions of NASA astronauts. The Report also discusses the legal, ethical and regulatory issues associated with using information on genetic susceptibility and prior radiation exposures to predict an astronaut's lifetime radiation-related health risks. This Report was prepared by Scientific Committee 1-13 on the Potential Impact of Individual Susceptibility and Previous Radiation Exposure on Radiation Risk for Astronauts. See Bibliography for information on the Report, including how to obtain a copy.
Bibliography Type:	Description: (Last Updated: 11/13/2019)
NASA Technical Documents	National Council on Radiation Protection and Measurements Scientific Committee 1-13. "Report No. 167 - Potential Impact of Individual Genetic Susceptibility and Previous Radiation Exposure on Radiation Risk for Astronauts." Bethesda, Md. : National Council on Radiation Protection and Measurements, 2011. (NCRP report ; no. 167) https://ncrponline.org/shop/reports/report-no-167-potential-impact-of-individual-genetic-susceptibility-and-previous-radiation-exposure-on-radiation-risk-for-astronauts/ ; https://ncrponline.org/publications/reports/ncrp-report-167/ , Jan-2011